

Comprehensive analysis of the influence of door width on the passenger flow time on Korean urban railways

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Abstract

This paper deals with the effect of door width on the passenger flow time on Korean urban railways using analysis of covariance (ANCOVA) and regression analysis. All regression coefficients are statistically significant in both analyses. The ANCOVA results show that increasing the door width can be quite useful in reducing passenger flow time. Regression analysis, which controls for other independent variables, also shows a clear relationship between door width and passenger flow time. In an additional regression model, we consider the interaction between door width and the number of boarding passengers. This paper also compares flow times estimated through two different methods: laboratory experiments and live railway observation, and finds no significant difference between the two approaches.

Keywords

Railway station, scheduling, train driving, railway transport, urban transport

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Introduction

Passenger flow time can be defined as the time passengers take to board and alight during a train stop at a station. The dwell time consists of three parts: passenger flow time, the time taken between the last passenger movement and the doors starting to close, and the time taken between completion of door closure and the start of wheel rotation. Some authors have described the latter two elements as the function time.¹ Both the passenger flow time and the dwell time are known to be influenced by many factors. In particular, the passenger flow time is directly influenced by passenger behavior. Therefore, a railway company should carefully determine the dwell time, so as to accommodate boarding and alighting passengers.

The influencing factors can be classified into three categories: facility, traffic, and rolling stock. The facility category includes horizontal and vertical gaps between a rail vehicle and a platform, as well as platform width. Most of the factors are defined as technical standards or guidelines for the design and construction of a railway. For example, a railway engineer, when he (or she) designs a platform, should calculate the platform width to avoid crowding based on a guideline level of service.

The traffic category includes the numbers of boarding/alighting passengers, a level of in-vehicle crowding, and so forth. The factors relate to the passengers

themselves, such as their volume, direction, density and even attitude. Therefore, other factors in the facility and rolling stock categories impose certain constraints or conditions.

Finally, the rolling stock category includes door width, as well as the existence of steps on a rail vehicle. Usually, the factors in the rolling stock category are not defined as public technical standards; rather, they evolved from the manufacturer's decisions about the rail vehicle. Thus, Korean rail vehicle manufacturers need to know the influence of factors on passenger behavior.

This paper focuses on the influence of door width on passenger flow time on urban railways. In most urban railways, the gaps between the rail vehicles and platforms are negligible, so door width becomes the significant remaining factor.

There have been several studies on the influence of door width on passenger flow time. Parkinson and Fisher² analyzed the passenger flow time for eight North American transit systems: BART, NYCT,

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PATH, Portland, San Diego, SF Muni, Vancouver, and TTC. Their report is known as TCRP-13. The report comments: "a variety of statistical analyses failed to show any meaningful relationship between door width and flow time". This was despite the fact that railway staff in the field believed the influence to be quite clear.

Heinz³ commented: "for a certain minimum door width (about 80–90 cm), increasing the width results in only a moderate increase in flow". The reason given is that the entire width of the door is not used if passengers form a queue. The trains with 80–90 cm doors tend to operate on regional or inter-city routes, and most of their vertical gaps are between 57 and 67 cm in Heinz's data. The considerable size of these vertical gaps can be another underlying reason for a moderate increase in flow rate with respect to door width. Heinz's conclusion may be most applicable to urban railway operation, where the vertical gaps are mostly very small, whereas door widths are generally much larger than 80–90 cm (e.g. over 130 cm on Korean urban railways).

Harris and Anderson¹ expressed a more generalized opinion than Heinz.³ Their opinion was that: "flow rates are not linear with door width, but there are quantum gains in capacity when widths increase to permit another passenger movement to take place simultaneously". However, they did not present proof of that opinion in their article.

Harris et al.⁴ commented: "many of the variables involved in determining movement rates have threshold values". Doors gain valuable capacity at widths (typically 1.4 m) that enable two simultaneous movements with ease. However, in practice, at least on the Korean urban railway, passengers do not usually wait in queues for the whole process of boarding or alighting. In most cases, queues are sustained only for the waiting period and when passengers first begin to board or alight, however, they break up after that period. Thus, formulating a flow time model based on the number of queues will probably not result in a good choice.

Wiggenraad⁵ claimed: "the relationship between the width of the passageway and the alighting/boarding time per passenger is clear". Wider doors led to 10% shorter times and narrower doors to 10% longer times. The claim was based on observations drawn from seven Dutch railway stations (four local and three inter-city stations). Staff at each station observed the boarding and alighting times and movements of passengers, using stopwatches and digitizers. However, their results on the influence of door width on passenger flow time were not analyzed using quantitative methodologies such as regression analysis.

The discussed literature has used live data collected from observations of railway passengers. In contrast, a few studies have collected data from artificially designed laboratory experiments to capture the influence of door width on passenger flow time. This

approach has the advantage of being able to generate more comprehensive data than the live field observation approaches, since it can examine a number of hypothetical boarding/alighting scenarios based on many possible door widths. Table 1 shows a list of research with data collected from both experiments and observations.^{1–3,5–14}

Rowe and Tyler⁷ presented results of psychophysical experiments jointly designed by Arup and UCL. They conducted the experiments at the UCL Pedestrian Access Laboratory. The experiment involved 125 participants, each performing boarding and alighting tasks for a 5 day period, with physical parameters and psychological stimuli being adjusted throughout the experiment to test their effect on performance. The experiment utilized video technology to capture the boarding and alighting sessions and these were analyzed in detail using behavioral observation software. They found that flow rates (of passengers) for boarding and alighting increased in a nonlinear fashion with respect to the door width (1.3, 1.5 and 1.8 m). Despite their well-organized experiments, they did not present a model of the relationship between passenger flow time and the door width.

Oh et al.⁶ presented the results of an experiment to forecast passenger flow time and to decide door width for the Ui LRT that is projected to open in Seoul in 2016. The Seoul city government requested them to verify whether the Ui LRT vehicle, which has two 1.5 m doors on each side, could maintain a dwell time of 18 s or less at the busiest station. All existing Seoul urban railways have four 1.3 m doors on each side. Thus, their doors are greater in number but narrower in total width than the doors of the Ui LRT vehicle. A laboratory experiment was designed to answer the question posed by the city government. The Gimhae LRT was selected as a test bed for the experiment, as its facilities were similar to those of the Ui LRT. Hired test attendees repeated boarding/alighting processes to create data. The results of statistical analyses proved that two 1.5 m doors could sufficiently handle the maximum number of passenger at the busiest station.

This paper presents two models of passenger flow time, with a focus on the influence of door width. The first is an analysis of covariance (ANCOVA) model, and the second one is a regression analysis model. There exist some advantages and limitations on the ANCOVA and regression analysis approaches. Among the advantages of ANCOVA is its ability to analyze the influence of categorical variables separately from the influence of other covariates. Therefore, ANCOVA is a good choice for intensive research to analyze the influence of door width, as in this paper. An advantage of the regression analysis approach is its ability to estimate a larger variety of variable-combinations that consist of either categorical or continuous variables than ANCOVA.

Table 1. Classification of state-of-the-art research on dwell times and passenger flow times.

Articles	Data collection	Rail systems	Methodologies	Aims of analyses	Door width	City/Country
Oh et al. ⁶	EX	UR	One-way ANOVA	Validation of 18 s for U _i LRT	Fixed to 1.5 m	Seoul/Korea
Rowe and Tyler ⁷	EX	UR	Statistical analyses on flow rate	Validation of 27 s. for 50 passengers	Successful DW = 1.8 m	Thames Link/UK
Douglas ⁸	Both	SR	Regression-based DT estimation	Calculate train dwell time for alternative layout of line and operating pattern	Considered	Sydney/Australia
Harris ⁹	EX	SR	LUL model-based FT estimation	Validation of rail vehicle designs for refurbishment	Fixed	London/UK
Puong ¹⁰	OB	UR	Regression model-based DT estimation	Present a regression model for MBTA-Red Line	Not considered	Boston/USA
Lam et al. ¹¹	OB	UR	Regression model-based DT estimation	Present a discomfort model that depends on platform crowding	Not considered	Hong Kong/China
Hermant ¹²	OB	RE	Statistical analyses on video-taped OB data	Present various features of passenger movement	Not considered	Cape Town/SA
Harris et al. ¹	OB	UR	Statistical comparison on DT	International comparison on DT	Nonlinear dependence	International
Heinz ³	OB	UR/RE/IC	Statistical reviews on FT	Describe effect of factors on FT	Moderate effect	Stockholm/Sweden
Wiggenraad ⁵	OB	UR/IC	Statistical surveys on FT/DT	Describe effect of factors on FT/DT	Wider door makes 10% shorter FT	Multiple/NL
Jong et al. ¹³	OB	UR/IC	Regression model on FT	Present regression models for UR and IC	Considered	Taipei/Taiwan
Parkinson and Fisher ²	OB	UR	Statistical analyses on FT/DT	Provide standard analysis schemes for station dwells	Statistical relationship not found	North America
Jiang et al. ¹⁴	OB	UR	Statistical surveys on DT	Present a simulation model to estimate delays in dwell	Not considered	Shanghai/China

Note: EX – experiment data, OB – observation data, LUL – London Underground Ltd, UR – urban railway, RE – regional railway, IC – inter-city railway, SR – suburban, FT – flow time, DT – dwell time.

Table 2. Data collection for Experiment A.

Door width(m)	Number of passengers				Number of samples	Passenger flow time (s)		
	Alighting	Standing	Boarding	Total		Mean	Std.	Individual
1.5	0	0	43	43	3	15.87	1.09	0.37
1.5	9	25	9	43	5	9.89	1.16	0.55
1.5	43	0	0	43	5	15.73	0.98	0.37
1.5	28	44	0	72	5	12.39	1.29	0.44
1.5	15	42	15	72	10	12.69	1.47	0.42
1.5	0	44	28	72	10	12.58	0.83	0.45

However, the regression analysis is unable to analyze the sole influence of categorical variables, as is possible in ANCOVA.

The ANCOVA model is used to quantitatively show the influence of door width on passenger flow time, by controlling for other independent variables, such as the numbers of boarding, alighting, or standing passengers. On the contrary, the regression model is used to show the collective influence of all independent variables on the passenger flow time.

Interestingly, Douglas⁸ commented that the passenger flow rate suggested by laboratory experiments is 30% smaller than the one calculated from live observations. His report attributed this to the possibility that test attendees behave less aggressively than actual railway passengers. The report presented a regression model after adjustment of the experimental data.

This paper compares data collected from laboratory experiments with data based on live observation. We found no significant difference between the two as far as Korean urban railways were concerned. We present a discussion of the differences in our results with those of Douglas.⁸

This paper is organized as follows. The next presents a comprehensive analysis on the influence of door width on passenger flow time. ANCOVA and regression analysis are performed and their results are presented. The section “Comparison between experimental and live observation data” compares data collected from laboratory experiments with those obtained from live observation. Finally, conclusions are drawn.

Influence of door width on passenger flow time

Collecting experimental data

The Korea Railroad Research Institute performed an experiment (Experiment B) in addition to that performed by Oh et al.⁶ (Experiment A). Experiment B aimed to capture the effect of variations in door width on passenger flow times. This section presents the results of ANCOVA and regression analysis using both data sets.

Experiment A aimed at verifying whether an 18 s flow time is enough for the Ui LRT line, planned to open in Seoul in 2016, with a 1.5 m door width. Table 2 shows a summary of data obtained in Experiment A. We treated the numbers of alighting, standing, and boarding passengers as factors that influence the passenger flow time. The passenger flow time, which is the response variable, is presented as the mean and standard deviation by simple calculation of the number of samples. In addition, the values of “Individual” are calculated dividing “Mean” by “Total” values. The assumption was made that most of the alighting passengers move first; boarding passengers move later without simultaneous movement in Experiment A. This assumption plays the role of the worst-case condition in the verification of the 1.5 m door width. The total number of test attendees was the sum of the boarding, alighting and standing passengers.

For Experiment B, a space measuring 4800 mm × 2920 mm (the same width as existing Seoul urban rail vehicles, but one-quarter the length), was prepared. The space was partitioned using panels. The length of the space was equal to the coverage length for a single door; due to rail vehicle having four doors on either side of every car (Figure 1).

Table 3 shows the values of factors, and the means and standard deviations of the response variable for Experiment B. The factors considered in Experiment B were identical with those of Experiment A, with the exception of the variation of the door width. The numbers of boarding and alighting passengers were calculated for 100, 150 and 200% in-vehicle congestion rates (measured relative to the normal capacity of the car). The door-width took on progressively increasing values: 1.3, 1.6, 1.8 and 2.0 m.

In both experiments, the data was collected under Korean urban railway conditions, in which any vertical and horizontal gaps between platforms and doors of rail vehicle are immaterial. Furthermore, the experiments assumed the absence of platform congestion. This assumption possibly reduces the passenger flow time; however, it was not possible to prepare congested platforms in time for the experiments.

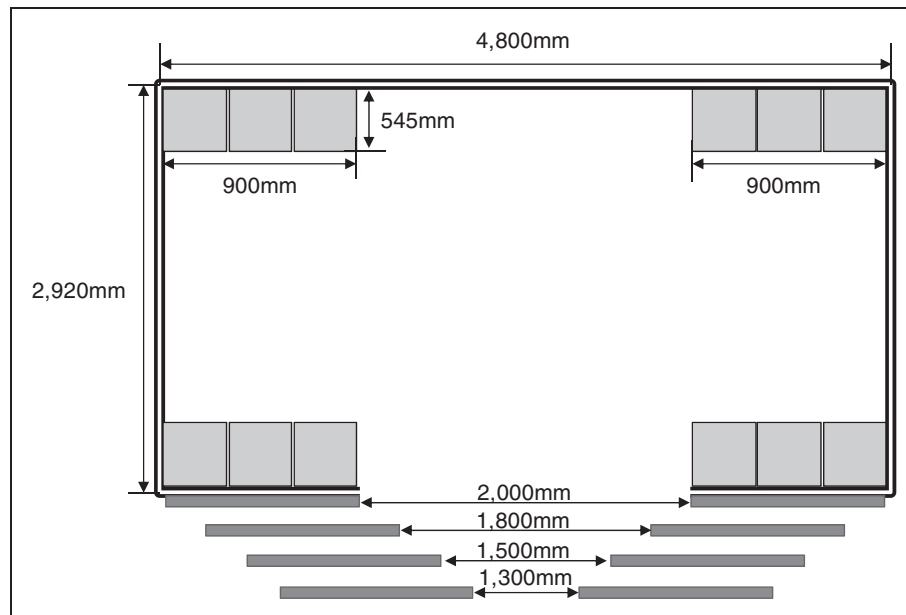


Figure 1. The layout of Experiment B.

Table 3. Data collection from Experiment B.

Door width (m)	Number of passengers				Number of samples	Passenger flow time (s)		
	Alighting	Standing	Boarding	Total		Mean	Std.	Individual
1.3	10	36	10	56	10	15.02	1.26	0.75
1.3	27	36	27	90	10	24.06	2.38	0.45
1.3	38	54	38	130	10	42.28	2.96	0.56
1.3	44	73	44	161	11	52.05	4.82	0.59
1.8	44	73	44	161	10	42.44	2.70	0.48
2.0	27	36	27	90	5	27.84	1.42	0.52
2.0	38	54	38	130	10	30.32	2.33	0.40
2.0	33	73	33	139	5	29.70	2.74	0.45
2.0	44	73	44	161	10	41.12	2.71	0.47

Table 4. ANCOVA for passenger flow times.

Variable	Sum of square	DF	Squared mean	F-value	p-value
Model	22,310.00	6	3718.33	290.87	0.000
Boarding	2845.52	1	2845.52	222.59	0.000
Alighting	2581.32	1	2581.32	201.93	0.000
Standing	289.37	1	289.37	22.64	0.000
Door width	1271.09	3	423.70	33.14	0.000
Error	1444.53	113	12.78		
Total	23,754.53	119			

ANCOVA for the passenger flow time

An ANCOVA was performed to find the influence of door width on passenger flow time, controlling for other covariates, such as the numbers of boarding,

alighting and standing passengers. Combining data from the two experiments, the categorical variable (door width) took on values of 1.3, 1.5, 1.8 and 2.0 m.

Table 4 shows the ANCOVA results. The ANCOVA model fits the data well ($R^2 = 0.944$).

Table 5. Estimation of regression coefficient in the ANCOVA.

Variables		Coefficient	Std. error	t-value	p-value
Covariate	Constant	0.286	1.429	0.200	0.842
	Boarding	0.458	0.031	14.920	0.000
	Alighting	0.460	0.032	14.210	0.000
	Standing	0.117	0.025	4.758	0.000
Level of factor	DW (1.5 m)	-5.062	1.006	-5.030	0.000
	DW (1.8 m)	-6.789	1.397	-4.859	0.000
	DW (2.0 m)	-8.242	0.909	-9.066	0.000

Table 6. Estimation of regression coefficients.

Variable	OLS estimation			LAD estimation		
	Coefficient	Std. error	p-value	Coefficient	Std. error	p-value
Constant	13.516	1.9563	0.000	15.870	3.8236	0.000
Boarding	0.478	0.0303	0.000	0.511	0.0409	0.000
Alighting	0.496	0.0297	0.000	0.499	0.0267	0.000
Standing	0.124	0.0247	0.000	0.118	0.0210	0.000
DW	-1.228	0.1319	0.000	-1.457	0.2561	0.000

All the covariates and factors are statistically significant at the 0.1% level.

Table 5 shows the estimated regression coefficients for the ANCOVA. The door width of 1.3 m plays the role of a reference category. Table 5 illustrates that an increase in the door width from 1.3 to 1.5 m reduces the passenger flow time by 5.06 s, whereas increasing it from 1.3 to 1.8 m and 2.0 m reduces the passenger flow time by 6.79 s and 8.24 s, respectively.

From Table 5, it is clear that increasing the door width considerably reduces the passenger flow time on the Korean urban railway system. In Table 5, the regression coefficients for the level of factor are conceptually similar to “threshold values” in Harris and Anderson.¹

In order to generalize the influence of door width on passenger flow time, a regression analysis was performed. The regression analysis also investigated the collective influence of the covariates on passenger flow time.

A regression analysis on the passenger flow time

For intuitive interpretation, we assume a simple linear polynomial regression model. In model (1), FT represents flow time, whereas B , A , S and DW represent the number of boarding, alighting, and standing passengers, and door width, respectively. The values of β are the regression coefficients of the corresponding independent variables and β_0 represents a constant term. DW was treated as a continuous variable that can

have a value between 1.3 and 2.0 m. This is different from ANCOVA, where door width is a categorical variable

$$FT = \beta_0 + \beta_1 \times B + \beta_2 \times A + \beta_3 \times S + \beta_4 \times DW \quad (1)$$

Table 6 shows the results of the regression analysis performed using two different estimation methods, ordinary least square (OLS) and least absolute deviations (LAD). The standard errors of the LAD coefficients were calculated using bootstrapping based on 200 repetitions. The OLS estimator required that the underlying errors of regression model be normally distributed random variables. However, the LAD estimator is robust against such distributional assumptions.

Both the OLS and LAD estimates are statistically significant at the 0.1% significance level. The OLS results in Table 6 show that the passenger flow time falls by 1.228 s for each 10 cm increase in door width. The corresponding reduction for the LAD results is 1.457 s. The two approaches did not yield significantly different results.

It is possible to conjecture that the response of passenger flow time to door width might be different when the number of boarding and/or alighting passengers is relatively high. This can be captured by including an interaction term between door width and the number of boarding and alighting passengers as an independent variable, as in regression

model (2). The last term in the model is the interaction term

$$FT = \beta_0 + \beta_1 \times B + \beta_2 \times A + \beta_3 \times S + \beta_4 \times DW + \beta_5 \times B \times DW \quad (2)$$

Table 7 shows the OLS estimation of regression coefficients for model (2). The model closely fits the data ($R^2 = 0.937$). The model is statistically significant since its F -value is 338.8, and the corresponding p -value is less than 0.01.

Table 7. Estimation of regression coefficient including an interaction term.

Variables	Coefficients	Std. error	t-value	p-value
Constant	-2.269	8.2431	-0.275	0.784
Boarding	0.913	0.2232	4.092	0.000
Alighting	0.492	0.0294	16.738	0.000
Standing	0.132	0.0247	5.343	0.000
DW	-0.162	0.5562	-0.292	0.771
Boarding $\times$ DW	-0.030	0.0150	-1.970	0.051

The p -value of the interaction term is 0.051, which is statistically significant. Even though DW is not statistically significant, with a p -value of 0.771, the hypothesis that the partial regression coefficients for both the DW and interaction terms are simultaneously zero is successfully rejected by a test, since its F -value is 46.346 ($DF1 = 2$, $DF2 = 114$), and the corresponding p -value is calculated to less than 0.01. Therefore, door width has a statistically significant influence if we include both its direct effects and its effects via its interaction with the number of boarding passengers.

Taking the interaction term into consideration, the influence of a 10 cm increase of door width on passenger flow time is estimated to be: $-(0.162 + 0.03 \times B)$. For example, a 10 cm increase of DW generates a reduction of 0.462 s ($= -0.162 - 0.03 \times 10$) in the passenger flow time at $B = 10$, however, it generates a larger reduction of 1.062 s ($= -0.162 - 0.03 \times 30$) at $B = 30$.

The estimation indicates that the reduction in passenger flow time achieved by increasing the door width is higher when the number of boarding passengers is relatively high Figure 2 shows the influence of door width on passenger flow time once the interaction term is taken into account.

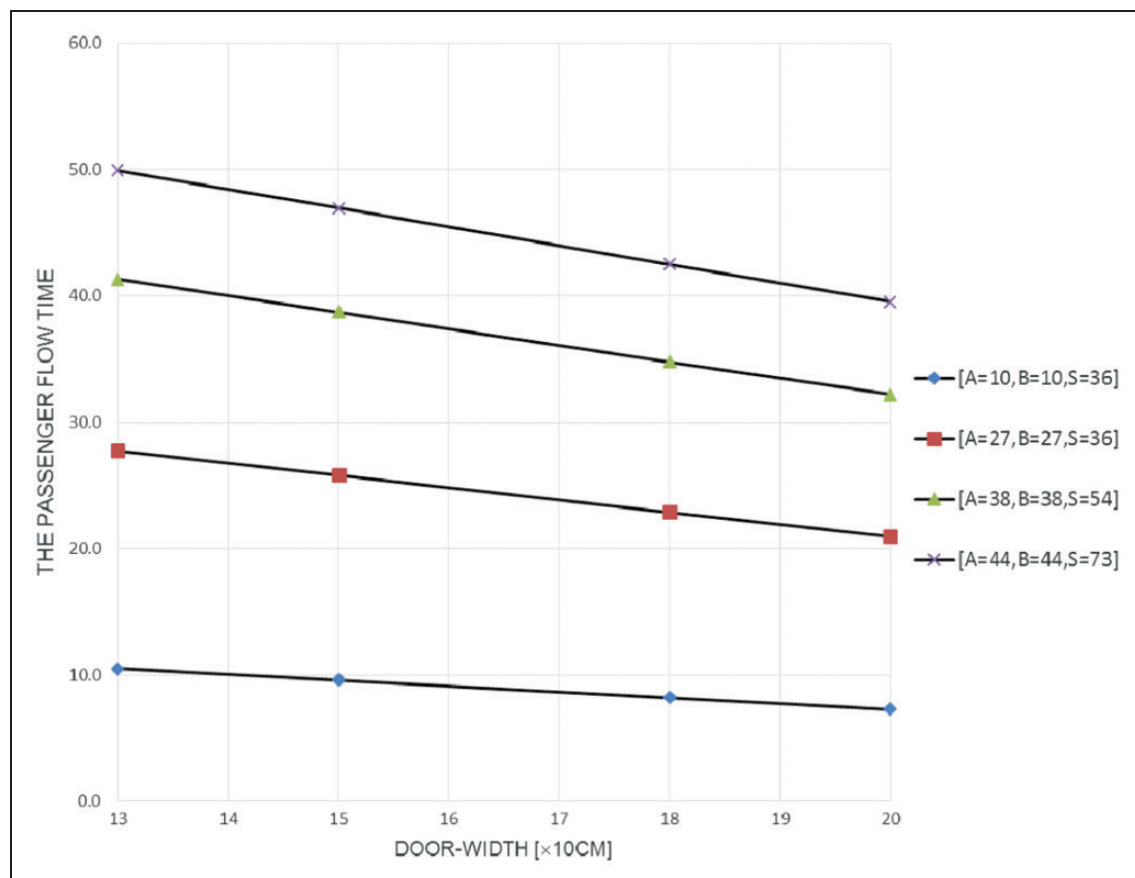


Figure 2. Influence of the door width considering the interaction term.

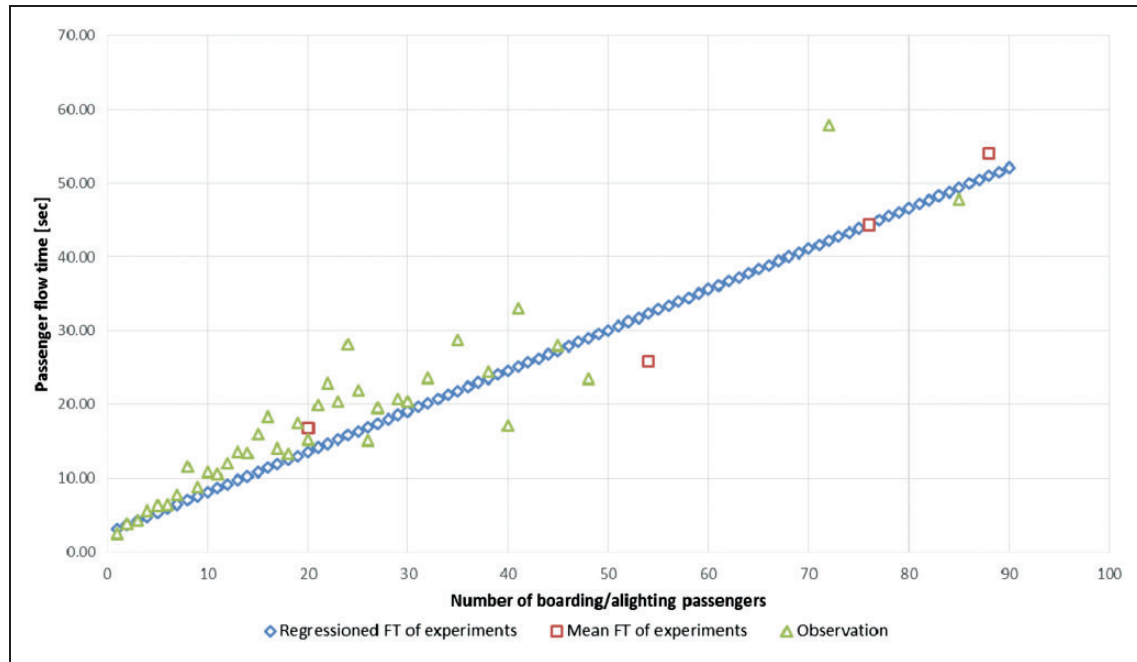


Figure 3. Comparison of FTs in experiments and observations.

Comparison between experimental and live observation data

Daamen et al.¹⁵ commented on an issue concerning real-world observations versus laboratory experiments in their research on pedestrian walkways. Despite a thorough preparation, some conditions affecting pedestrian flows were significantly different from the researcher's expectation. A laboratory experiment can provide copious amounts of information and data; however, it has disadvantages in that the experimental environment and the attendees' attitudes lead to different results from the real-world boarding/alighting process.

This paper compares the passenger flow time calculated based on experimental data (from Experiments A and B) with passenger flow times calculated using live observations. By doing so, we try to check the comments of Douglas⁸ on the difference between the two data sources.

Our data observations were made on the Seoul Metro Line 2 and the KORAIL Bundang Line. Both these lines are located in the Korean capital area, where all rail vehicles have 1.3 m doors. We collected 294 data samples from the two lines, and made a selection of 38 samples, in which boarding and alighting passengers were greater than or equal to 20 passengers. This selection was made to create the same conditions as the 41 experiment data samples of 1.3 m door width from Table 3.

We performed a simple regression analysis (model (3)) using the experimental data. The goodness of fit

was not bad ($R^2 = 0.89$) and β_1 was estimated to be statistically significant ($p = 0.00$)

$$FT = \beta_0 + \beta_1 \times (B + A) \quad (3)$$

The results of the regression analysis are plotted in Figure 3 together with the mean values of the experimental data for each boarding/alighting scenario. The observed data are plotted in Figure 3, in addition to the experimental data. The observed data are balanced on either side of the regression line and do not significantly deviate from it. This indicates a lack of significant difference between the experimental and observation data sets for Korean urban railways.

Thus, our conclusion is different from Douglas.⁸ Actually, the researchers who performed Experiments A and B found that the test attendees conformed to a standardized pattern of boarding or alighting processes after enough repetitions. This may possibly reduce the passenger flow time in our laboratory experiments. This also marks a difference in test attendants' behavioral attitude between our work and that of Douglas.⁸

Conclusions

An ANCOVA and a regression analysis were performed to analyse the influence of door width on passenger flow time. Two separately collected data sets (from Experiments A and B) were used for the analyses. Most of the conditions under which the data were collected correspond to the actual conditions on the Korean urban railway system. The ANCOVA results show that an increase in door

width can be quite useful in reducing passenger flow time. The results agree with previous research in terms of the general features on the influence of door width.

Moreover, two types of regression models were presented to examine the relationship between the number of boarding, alighting or standing passengers, the door width and the passenger flow time. One includes an interaction term between door width and the number of boarding passengers as an independent variable, whereas the other does not include any interaction between these variables.

Finally, we use data sets collected from the Korean urban railway to review the issue of differences between data generated from laboratory experiments and live railway data. In contrast with Douglas⁸, we find no significant differences between experimental and observation data for the Korean urban railway. Douglas⁸ explained his findings in terms of more aggressive behavior by actual railway passengers. However, our work suggests that test attendees' behaviors may possibly become standardized, affecting the estimate of passenger flow time. Future research into the behavior of live railway passengers as well as test attendees is required.

Declaration of Conflicting Interests

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